

Document Purpose

Note on Friday 14-Aug customer Review of Broke System export

Referring to the files in Salis's message

<https://ensembleanalyticsteam.slack.com/archives/C0B4D92FE30/p1786644000849329>

Loom

2 Loom to support this

Part 1 <https://www.loom.com/share/aff86a0925474a8b904a2f241deb334a>

Part 2 <https://www.loom.com/share/4dce4ad5a0564c119e2d0a607bd4d087>

Note.

You can search PDFs by text. This is really useful to be able to file icons in the pdf diagram

Abbreviations

Problem. Not all Functional Location Description words have been abbreviated when abbreviations already exist.

File: Broke System.functional_locations.xlsx

Column: Functional Location Description

Cell F15

Cell Value: 35-24L004 BROKE CONVEYOR 2

Feedback: CONVEYOR *must* be abbreviated to CVYR. There are several other examples too. Some words are already abbreviated but some are not.

Solution: You *must* use an abbreviation where there is one. Revisit the Abbreviation mapping logic. Use Regsex to extract every word in the Functional Location Description field, and abbreviate where an abbreviation already exists.

Empty Work Center column

File: Broke System.functional_locations.xlsx

Column: Work Center

Problem: the column is empty

Solution. Populate the column. Similar values to the equipment spreadsheet

Subordinate Equipment

File: Broke System.equipment

Column: Position

Problem: Superordinate Equipment Position must reset

See the screen shot below where I have corrected the position. See line 3 is subordinate equipment and its position must reset. And then line 4 continues from line 2. And line 5 is another subordinate equipment so its position resets.

Picture Subordinate equipment and another layer of hierarchy in the tree.

Solution.: Each new piece of subordinate equipment must reset the numbering

0	5001-PM03-BR-BR1-35-24L001	35-24LC-576		0040
1	5001-PM03-BR-BR1-35-24L001	35-24NC-577		0050
2	5001-PM03-BR-BR1-35-24L001	35-24-095		0060
3	5001-PM03-BR-BR1-35-24L001	35-24LV2-576	35-24-095	10
4	5001-PM03-BR-BR1-35-24L001	35-24-096		70
5	5001-PM03-BR-BR1-35-24L001	35-24-207	35-24-096	10
6	5001-PM03-BR-BR1-35-24L001	35-24-217	35-24-096	20
7	5001-PM03-BR-BR1-35-24L001	35-24-1105	35-24-096	30
8	5001-PM03-BR-BR1-35-24L001	35-24-093		80
9	5001-PM03-BR-BR1-35-24L001	35-24-234	35-24-093	10
0	5001-PM03-BR-BR1-35-24L001	35-24L001.1		90
1	5001-PM03-BR-BR1-35-24L001	35-24L001.2		100
2	5001-PM03-BR-BR1-35-24L002	35-24-002.1		0010
3	5001-PM03-BR-BR1-35-24L002	35-24XS-662		0020
4	5001-PM03-BR-BR1-35-24L002	35-24LC-622		0030
5	5001-PM03-BR-BR1-35-24L002	35-24GS-625		0040

Equipment Text – Line Identifiers

File: Broke System.equipment
Column: Equipment Text

Problem: All Line's Equipment text must start with LN.

Example 1

Cell F12 value is "35-24-095 PRESS PLPR PP-200 LINE".
But it should be "LN 35-24-095 PRESS PLPR PP-200"
The word "LINE" is removed but the text is prefixed with LN.

Example 2

Cell F14 value is "35-24-096 PRESS PLPR PP-900 LINE".
But it should be "LN 35-24-096 PRESS PLPR PP-900"
The word "LINE" is removed but the text is prefixed with LN.

Solution. Ensure all Line's Equipment Text are prefixed with LN and the word LINE does not appear in the text

Equipment Text – Valve Identifiers

File: Broke System.equipment
Column: Equipment Text

Problem: All Hand Valve Equipment text must start with HV

Example

Cell F95 value is "35-24HV-548 HAND VLV"
But it should be "HV 35-24-548 35-24L005 AV".
Note that the middle HV has been removed from the text.
Where 35-24L005 = the REEL PLPR, refer to Broke System.functional_locations.xlsx, cell F15
And AV is Automatic Value

See the value in the screen print below, you see it has a circle above it. Look at the diagram key and you'll see that it is an automatic valve! We will need to use Computer vision to make this determination!

LEGENDS:		INSTRUMENTS		PIPELINES	
<u>SYMBOLS:</u>			CONTROL ROOM INSTRUMENT WITH DCS CONTROL		PLANT AREA CODE
	HAND VALVE		CONTROL ROOM INSTRUMENT WITH MCS CONTROL		SUB-PROCESS CODE
	VALVE NORMALLY CLOSED		QUALITY CONTROL SYSTEM (QCS)		POSITION NUMBER
	CHECK VALVE		CONTROL ROOM INSTRUMENT		FLOW MEDIA
	PRESSURE REDUCING VALVE		LOCAL CONTROL PANEL INSTRUMENT		DN SIZE
	SAFETY VALVE		LOCALLY MOUNTED INSTRUMENT		PIPE CLASS
	FLUSHING		PLC CONTROL		
	SAMPLING				
	DRAINAGE				
	AUTOMATIC VALVE				
	AUTOMATIC VALVE, ELECTRICAL MOTOR ACTUATOR				
	SIGHT GLASS				
	ORIFICE PLATE				
	STEAM TRAP				

And looking at the key you will see it is a Drainage Valve that is normally closed

Example of an 35-24-137 Drainage Valve Normally Closed

